

Daily Science Articles

Topic: Why can't we multiply vectors?

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Why can't you multiply vectors?

What you might say is:

1. "Yes you can, and it's called the dot product."
2. "Yes you can, and it's called the cross product."
3. "What do you mean? I multiply vectors all the time."

If you're a shader programmer, you might be thinking of component-wise multiplication (the Hadamard product). Let's start with the basic anatomy of a vector:

$$\vec{v} = (v_x, v_y, v_z) = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

This is in an orthonormal basis. Now let's look at common products:

1. Dot product:

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z = \sum_{i=1}^3 a_i b_i$$

2. Cross product:

$$\vec{a} \times \vec{b} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix}$$

3. Hadamard (component-wise) product:

$$\vec{a} \circ \vec{b} = \begin{bmatrix} a_x b_x \\ a_y b_y \\ a_z b_z \end{bmatrix}$$

So we have the dot, cross, and Hadamard products. What about "the product"

$$\vec{a} \vec{b}$$

Addition, subtraction and scalar multiplication apply to vectors, but isn't the vector product just multiplying the corresponding terms together? Well, looking at our types of numbers, we start with natural numbers. Natural numbers are not closed under subtraction since we enter the integers, negative numbers (-3 for example). If we apply division we leave integers and enter rational numbers ($1/3$ for example). But if we apply exponents to rational numbers, then we enter the real numbers ($\sqrt{5}$ for example). But that too isn't closed yet, since we enter complex numbers ($\sqrt{-5}$ for example). A quick background about complex numbers, $i^2 = -1$ is the imaginary unit, doing math with it, its treated like some variable:

$$\begin{aligned} 2i(4i - 3) &= 8i^2 - 6i \\ &= -8 - 6i \end{aligned}$$

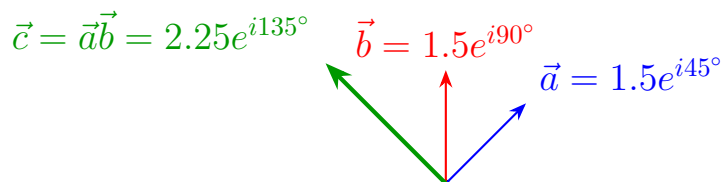
so we can think about the number having two parts, imaginary part ($z_i = -6$) and a real part ($z_r = -8$), both parts make up a complex number. We can also represent them as follows, a complex number z :

$$z = (z_r, z_i) \quad \text{looks similar to} \quad \vec{v} = (v_x, v_y)$$

so just like we draw 2d vectors we can do the same with complex numbers, but now if we multiply complex numbers together since they are similar to vectors so we now can know what vector product is, have a look:

$$\begin{aligned} ab &= (a_r + a_i i)(b_r + b_i i) \\ &= a_r b_r + a_i b_i ii + a_r b_i i + a_i b_r i \\ &= a_r b_r - a_i b_i + a_r b_i i + a_i b_r i \\ &= a_r b_r - a_i b_i + i(a_r b_i + a_i b_r) \end{aligned}$$

We also result in a complex number! the real part $a_r b_r - a_i b_i$ and the imaginary part $(a_r b_i + a_i b_r)$, if represented the 3 as vectors, $\vec{a}$, $\vec{b}$ and $\vec{a}\vec{b}$ and see the behavior for ourselves whats so magical?

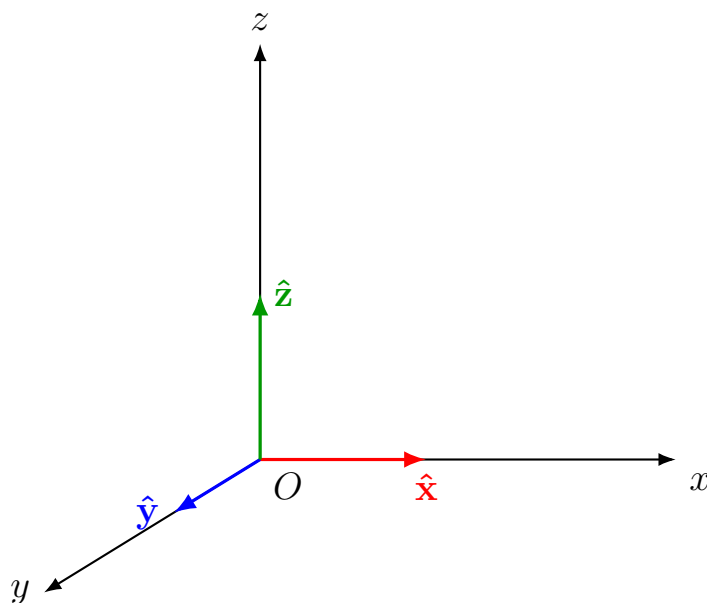


So basically the 2 angles of $\vec{a}$ and $\vec{b}$ are being added up and we have no clear relation, and most importantly what happens in 3d?

Complex numbers are closed under everything, but what if we step to 3 dimensional vectors, now rewriting vectors:

$$\vec{v} = (v_x, v_y, v_z) = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = v_x \hat{x} + v_y \hat{y} + v_z \hat{z}$$

where $\hat{x}, \hat{y}$ and $\hat{z}$ are the unit vectors of length one, they make up all vectors in 3d.



Component wise scalar multiplication applies, addition and subtraction also applies as we just work the similar terms together using simple algebraic manipulation. I assume now we have our tools to just

$$\begin{aligned} \vec{a}\vec{b} &= (a_x \hat{x} + a_y \hat{y} + a_z \hat{z})(b_x \hat{x} + b_y \hat{y} + b_z \hat{z}) \\ &= a_x b_x \hat{x}\hat{x} + a_x b_y \hat{x}\hat{y} + a_x b_z \hat{x}\hat{z} \\ &\quad + a_y b_x \hat{y}\hat{x} + a_y b_y \hat{y}\hat{y} + a_y b_z \hat{y}\hat{z} \\ &\quad + a_z b_x \hat{z}\hat{x} + a_z b_y \hat{z}\hat{y} + a_z b_z \hat{z}\hat{z} \end{aligned}$$

Well this thing is not a vector anymore, $(?)\hat{x} + (?)\hat{y} + (?)\hat{z}$

Vectors are not closed under multiplication.

That is the answer, you can not multiply vectors in 3d.

We are not going to give up that easy, we can have a look at some rule assumption which is vector magnitude:

$$\vec{v}^2 = ||\vec{v}||^2 = \left(\sqrt{v_x^2 + v_y^2 + v_z^2} \right)^2$$

so what about our basis vectors: $\hat{x} \hat{y} \hat{z}$

This will cleanly reduce to:

1. $\hat{x}\hat{x} = \hat{x}^2 = 1^2 = 1$
2. $\hat{y}\hat{y} = \hat{y}^2 = 1^2 = 1$
3. $\hat{z}\hat{z} = \hat{z}^2 = 1^2 = 1$

Now we can start simplifying and separating the terms:

$$\begin{aligned} \vec{a}\vec{b} &= a_x b_x + a_y b_y + a_z b_z \\ &\quad + a_y b_z \hat{y}\hat{z} + a_z b_y \hat{z}\hat{y} \\ &\quad + a_z b_x \hat{z}\hat{x} + a_x b_z \hat{x}\hat{z} \\ &\quad + a_x b_y \hat{x}\hat{y} + a_y b_x \hat{y}\hat{x} \end{aligned}$$

Now I will do some extra steps and I'll tell you to just watch:

$$\begin{aligned} \vec{v}^2 &= ||\vec{v}||^2 \\ (\hat{x} + \hat{y})^2 &= (\sqrt{2})^2 \\ \hat{x}\hat{x} + \hat{x}\hat{y} + \hat{y}\hat{x} + \hat{y}\hat{y} &= 2 \\ 1 + \hat{x}\hat{y} + \hat{y}\hat{x} + 1 &= 2 \\ \hat{x}\hat{y} + \hat{y}\hat{x} &= 0 \\ \hat{x}\hat{y} &= -\hat{y}\hat{x} \end{aligned}$$

You can apply what I did above knowing that these are basis vectors, going back to what we were doing:

$$\begin{aligned} \vec{a}\vec{b} &= a_x b_x + a_y b_y + a_z b_z \\ &\quad + \hat{y}\hat{z}(a_y b_z - a_z b_y) \\ &\quad + \hat{z}\hat{x}(a_z b_x - a_x b_z) \\ &\quad + \hat{x}\hat{y}(a_x b_y - a_y b_x) \end{aligned}$$

That is the result of some algebraic manipulation and factoring, but you immediately recognize the $a_x b_x + a_y b_y + a_z b_z$ as the dot product and the $\hat{y}\hat{z}(a_y b_z - a_z b_y) + \hat{z}\hat{x}(a_z b_x - a_x b_z) + \hat{x}\hat{y}(a_x b_y - a_y b_x)$ as the cross product.

We are still stuck with the $\hat{y}\hat{z}, \hat{z}\hat{x}, \hat{x}\hat{y}$, so we try some stuff like

$$(\hat{x}\hat{y})^2 = \hat{x}\hat{y}\hat{x}\hat{y} = -\hat{x}\hat{x}\hat{y}\hat{y} = -1 \cdot 1 = -1$$

that is useful, but wait, $i^2 = -1$, this applies to the rest:

$$(\hat{x}\hat{y})^2 = (\hat{z}\hat{x})^2 = (\hat{y}\hat{z})^2 = (\hat{x}\hat{y})(\hat{z}\hat{x})(\hat{y}\hat{z}) = -1$$

but we have seen this before in something called quaternions:

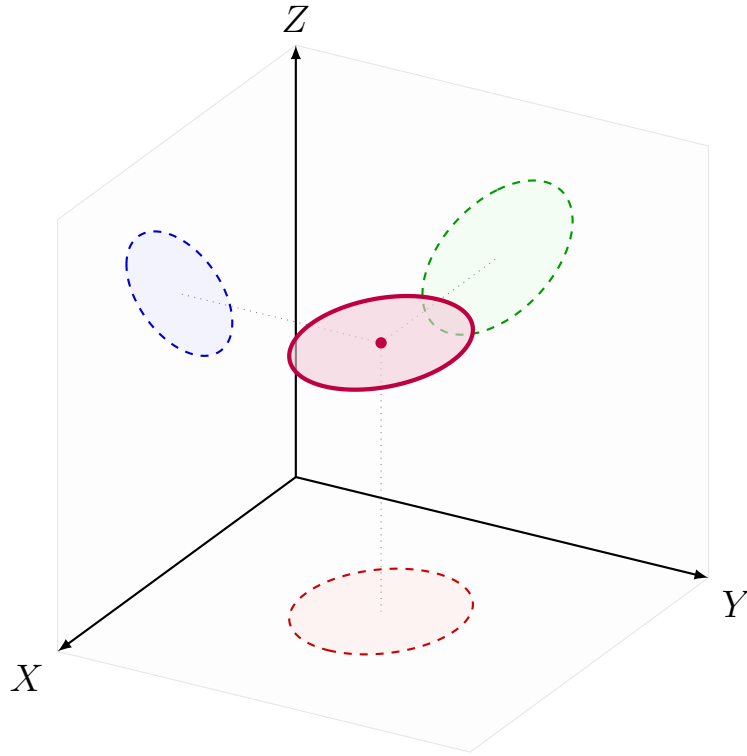
$$i^2 = j^2 = k^2 = ijk = -1$$

Quaternions are composed of $(1, \hat{y}\hat{z}, \hat{z}\hat{x}, \hat{x}\hat{y})$ and you take a look at this and you can match the terms

$$\begin{aligned} \vec{ab} &= a_x b_x + a_y b_y + a_z b_z \\ &\quad + \hat{y}\hat{z}(a_y b_z - a_z b_y) \\ &\quad + \hat{z}\hat{x}(a_z b_x - a_x b_z) \\ &\quad + \hat{x}\hat{y}(a_x b_y - a_y b_x) \end{aligned}$$

$$(a_x b_x + a_y b_y + a_z b_z, (a_y b_z - a_z b_y), (a_z b_x - a_x b_z), (a_x b_y - a_y b_x))$$

The basis vectors of Quaternions are like planes, they are like 2 vectors forming a plane, lets call them bi-vectors. Lets make it more intuitive, cut a small piece of paper with some shape (preferably a circle) and hold that circle in the corner of your room, turn off the lights and bring your phone flashlight. now hold the paper in some way in the air in that corner, and aim the phone light straight down on it, you will see a shadow below it, that is the shadow which resembles that bi-vector, which are 2 vectors that form that shadow, you can now test this at the other left and right walls by aiming the light towards them and casting the shadow onto them.



Ah, well this is a very good presentation yet its not very accurate, but you got the point. So this is like having some oriented plane/area casting shadows on these bi-vectors, each bi-vector has some coefficient.

$$(0.19, -0.25, 0.17) \iff (\hat{y}\hat{z}, \hat{z}\hat{x}, \hat{x}\hat{y})$$

Bi-vectors represent the minimum information required in any given dimension, to store both a plane and a magnitude. This is why they show up when dealing with rotations, because rotations happen in a plane. Just to clear another point that what we just compared above in terms of coefficients was the same coefficients of the cross product, but the cross product does not have bi-vectors, a wedge product has these bi-vectors (more on that, cross is more limited to 3d and 7d, while wedge works in all dimensions yet its harder to understand). The wedge product helps us generalize some rules regarding curvature in all dimensions.

What we just discussed is called geometric algebra, it is by far one of the greatest shit in math that I would probably come across as a 17 year old doing nerdy math research. *This article is 300 lines of Latex code.*

Enough talking, why can't we multiply vectors? Maybe because your teacher did not explain geometric algebra to you. Now you can multiply vectors, but the result is not clearly a vector but now you know what result that you would get, some plane in space casting shadows.